221-2327 Sec-E Activity #

1. Three levels of correctness

* **Syntatic Correctness:-**

This means the program follows the rules of programming languages. e.g There should be no mistake like missing brackets, semi-column, etc in code.

* **Semantic Correctness:-**

This means the program gives the correct output. even if code has no syntax errors, it can still be wrong if it doesnot do what the programmer intended.

* **Logical Correctness:-**

This means the program solves the problem in right way. The logic should match the real requirements. A program may run fine but if steps are wrong, the final result will still be incorrect.

2.

One Major lesson is that software errors can be very dangerous, especially in safety-critical systems. because of software faults, patients received massive radiation overdoses. So systems in health-care, aviation or other critical areas must be tested carefully.

3.

A testcase is a set of conditions, input, steps and expected results used to check ~~what~~ whether a system/feature is working well.

Common sections include:
Test Case ID, Test Case Name, Purpose, Pre-Condition, Steps, Data, Results Expectation, Actual result, status, remarks.

4. ID: TC_BANK_001
Name: Verify Money Transfer handling when network connection is lost after deduction.
Objective: To verify that the system handles a network failure if amount is deducted but confirmation is not shown.
Pre-Conditions: * Users is logged into online banking.
* Users have enough balance in Account
* Receiver account is valid.
* Internet connection availability initially
Test Steps: First login → Money transfer Page → Receiver Account information entering → transfer amount entering → Click on Transfer → Disconnect the network immediately after amount deduction before confirmation appears → Reconnect network → login again → Check Transaction history and balance of both account.
Expected Results:- The system should hande situation safely, The transaction should either completed successfully or refunded automatically, No loss of money, no duplicate transaction, clearly showing final status.